



Spiking Neural Networks for Text Classification using MSF and LIF Models

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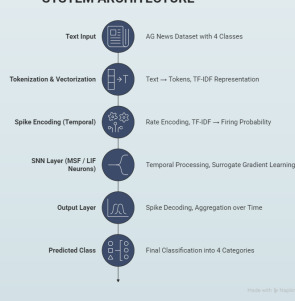
Problem Statement

- ANNs use continuous activations, high computational cost (MAC-heavy).
- SNNs are efficient but LIF-SNN limited to binary spikes (no intensity encoding)
- Key Gap: No model captures both instantaneous intensity + temporal dynamics
- MSF-SNN: Captures instantaneous rate + temporal patterns, bridges ANN $\leftrightarrow$ SNN

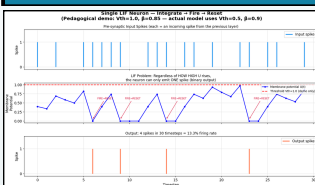
Feature	ANN (ReLU)	LIF-SNN	MSF-SNN
Neuron output	Continuous	Binary 0/1	0 to D
Energy ops	MAC 4.6 μ J	AC 0.9 μ J	AC 0.9 μ J
Intensity coding	✓	✗	✓
Temporal coding	✗	✓	✓
Extra params?	—	—	None

Methodology

SYSTEM ARCHITECTURE

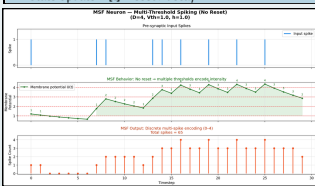


SNN Models



MSF Neuron Dynamics

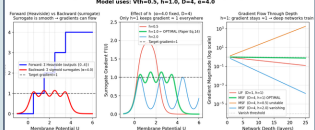
- Membrane Update: $U[t] = \beta U[t-1] + W \cdot \text{In}[t]$
- Spike Generation: $S[t] = \sum \alpha U[t] - d \cdot \text{vth}$
- State Update: $H[t] = \text{reset} / \text{decay}$



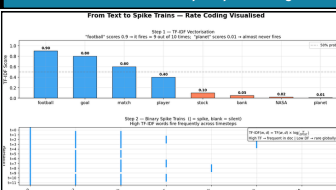
Surrogate gradient and Threshold

- Heaviside function, non-differentiable (no gradients)
- Use sigmoid surrogate (backward pass only)
- MSF advantage:
 - Sum of D surrogates $\rightarrow$ stable gradient flow
 - Optimal: $h = 1 \rightarrow$ gradient ≈ 1 across layers
 - Prevents vanishing/exploding gradients

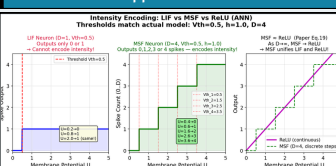
Surrogate Gradient - Why h=1 and not Any Optimal (Paper Eq.14)



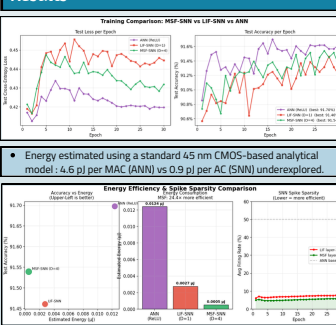
How text is converted into input spikes using TF-IDF



How MSF can be approximated to ANN



Results



SNN Implementation for IMAGE Classification



CIFAR-10 - T-SNE FEATURE CLUSTERS (MSF)



Conclusion

- Successfully implemented an SNN-based text classifier using TF-IDF + rate encoding on the AG News dataset using Surrogate Gradients.
- Demonstrates that SNNs can handle NLP tasks with competitive performance and energy-efficient potential.

Future Work

- Improve accuracy with advanced encodings (temporal/semantic embeddings)
- Deploy on neuromorphic hardware for real energy efficiency gains
- Explore recurrent SNN architectures.

References

- [1] L. Fan et al., "A multi-synaptic spiking neuron for simultaneously encoding spatiotemporal dynamics," Nature Communications, vol. 16, no. 1, Aug. 2025, doi: <https://doi.org/10.1038/s41467-025-62311-6>.
- [2] M. Horowitz, "1T Computing: the energy problem and what we can do about it," 2014 IEEE International Solid-State Circuits Conference Digest of Technical Papers (ISSCC), San Francisco, CA, USA, 2014, pp. 10-14, doi: [10.1109/VSSCC.2014.6757323](https://doi.org/10.1109/VSSCC.2014.6757323).

We would like to thank Prof. Manisha Padala for her unwavering support and guidance throughout the project.